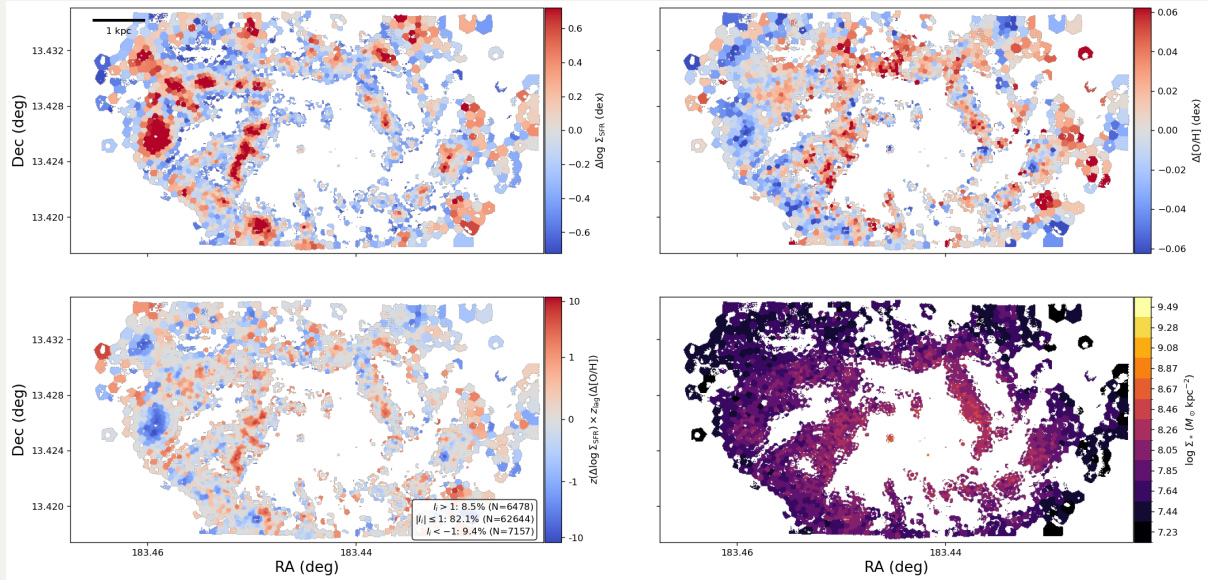
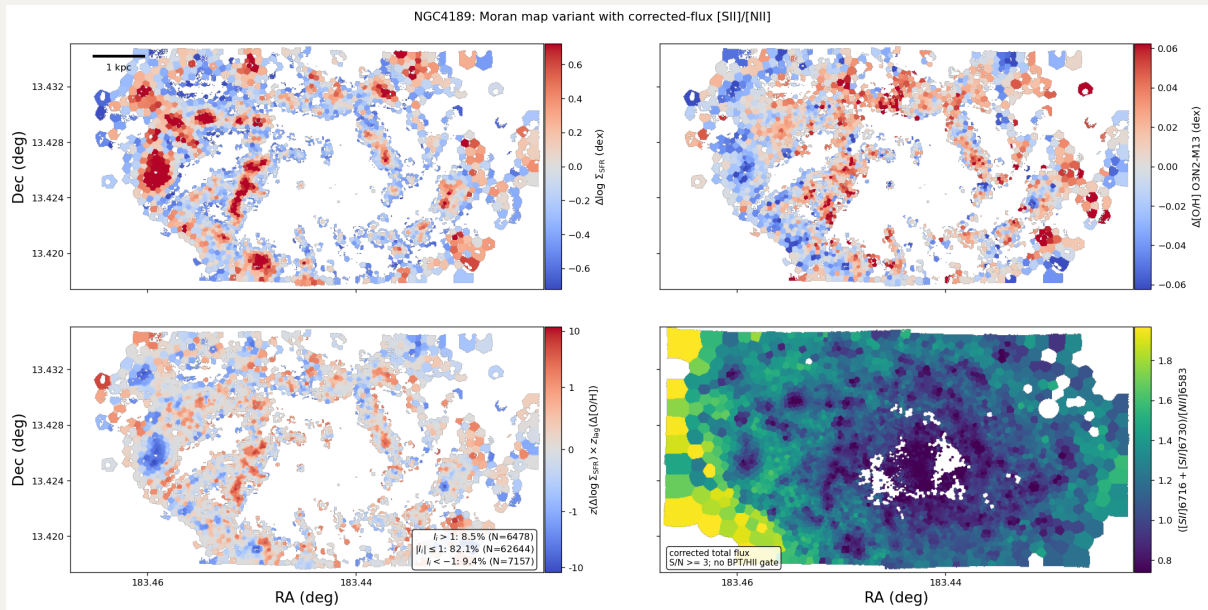


20260529 NGC4189 SII NII ratio

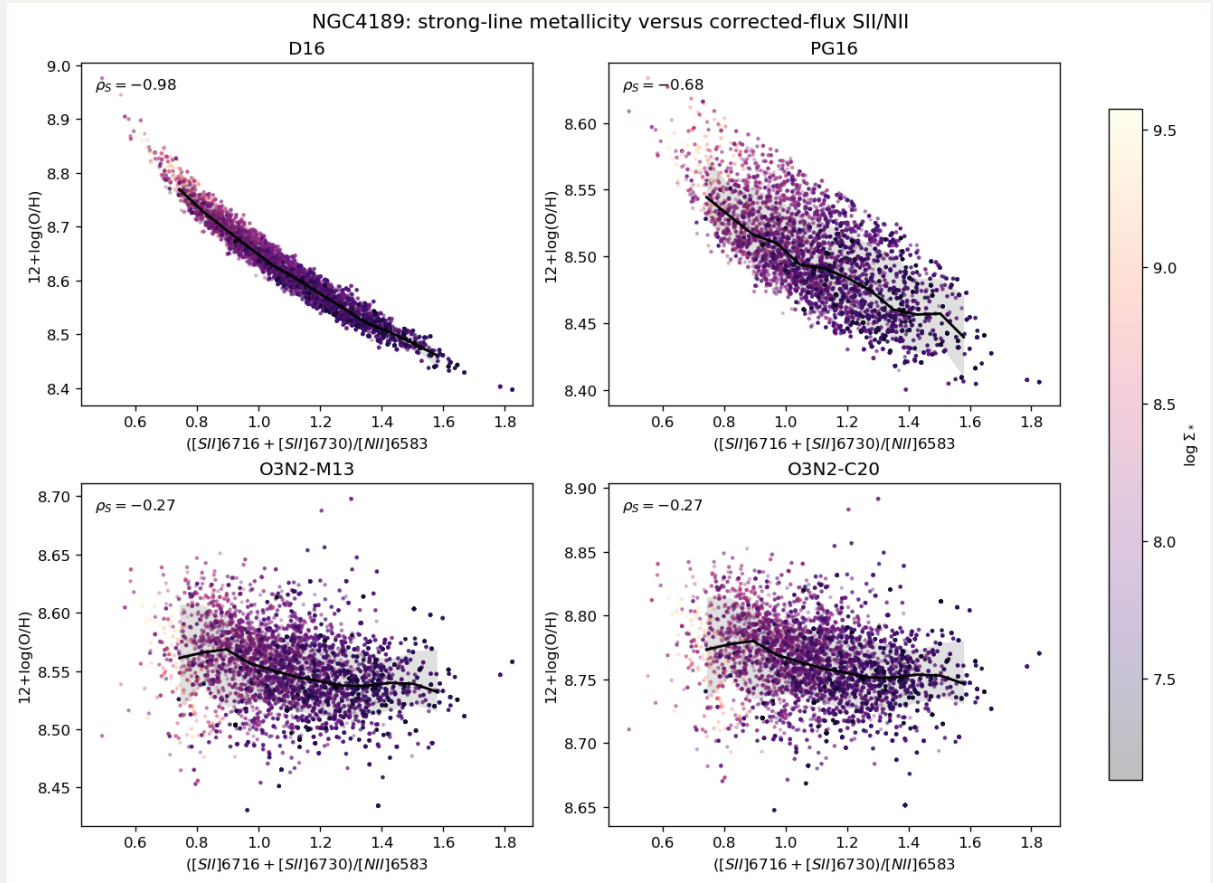
First we show the moran maps for NGC4189



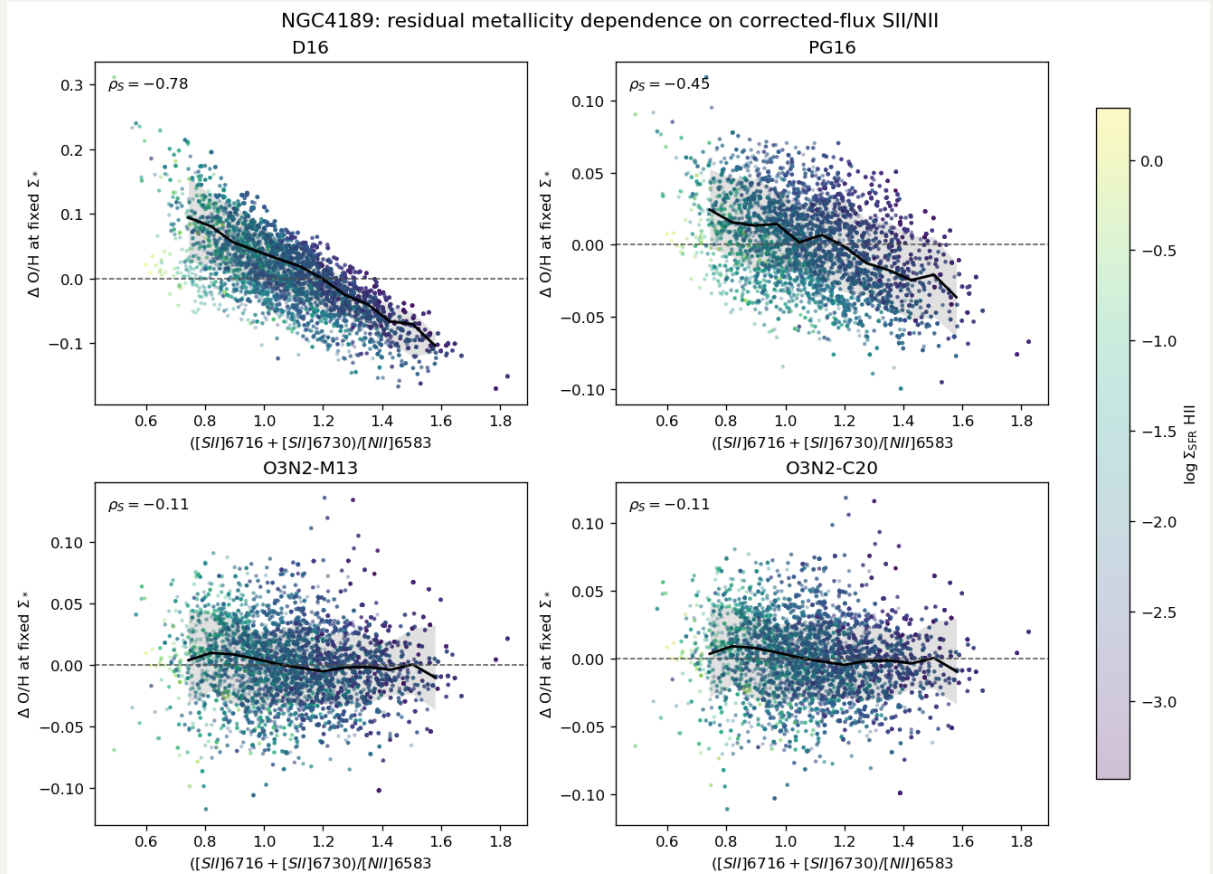
Then, we replace the bottom right panel with $[\text{S II}]/[\text{N II}]$ (note that not limited to HII spaxels) to check the HII region position. Overall, their position match.



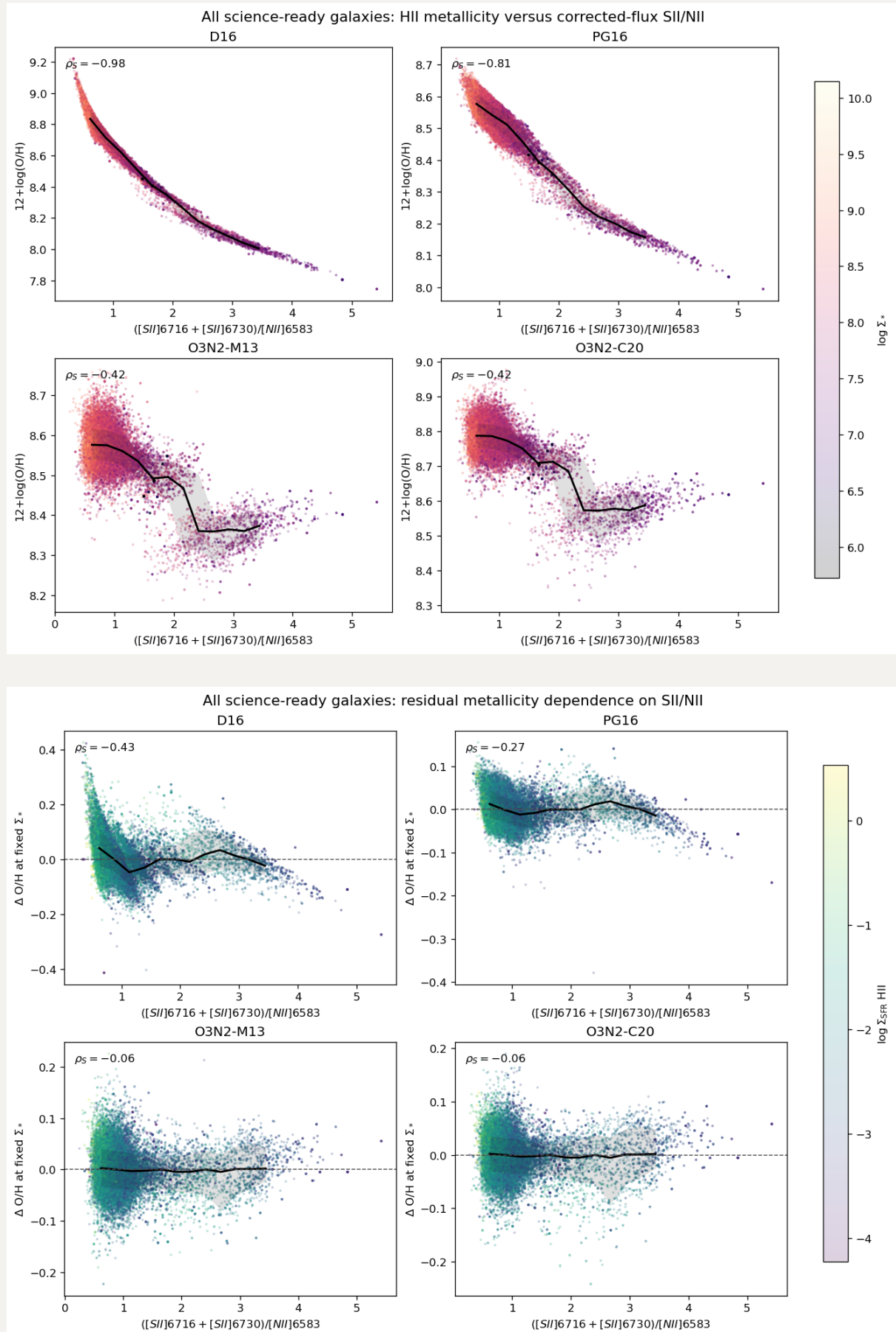
And we also check the the existing HII metallicity (D16, PG16, O3N2-M13, and O3N2-C20) versus $[\text{S II}]/[\text{N II}]$, colorcoded by stellar mass. And no wonder, D16 and PG16 show strong dependence on it as they are both using N2S2 indicators, while two O3N2 calibrations are insensitive to $[\text{S II}]/[\text{N II}]$.



We can also replace the y axis with $\Delta[\text{O}/\text{H}]$ at fixed stellar mass, and colocoloded by SFR.



Similarly, we can repeat these two plots with all 18 science-ready galaxies we have so far:



For fun, we can have a look at n_e and other T_e -based metallicities and clearly they are independent on [S II]/[N II] ratio.

